# Environment Setup

SQL Server and SSMS for the Data Warehouse project, on Windows and Mac

The guide gets you to the first line of the project: SQL Server installed, the six source files where the scripts look, the database created. With the fixes for the common errors.

## 1 Choose your setup

| Path | For | Same screens as the videos | Cost |
|---|---|---|---|
| **SQL Server Express + SSMS**, section 3 | Windows | yes | free |
| **Docker**, section 5 | Mac | the same SQL, in Visual Studio Code | free |
| **Parallels with Windows**, section 6 | Mac | yes, SSMS inside Windows | about 100 dollars a year, 14-day trial |

Read only your path. Windows: sections 3 and 4. Mac: section 5, or section 6 for Windows inside the Mac. Everyone: section 7 when something breaks.

My advice: on Windows, take the first path, it is the one in the videos. On a Mac, take Docker: free, light, and the SQL is the same. Take Parallels if you want SSMS exactly as in the videos and your Mac has 16 GB of memory.

## 2 What you get in the materials

Download the project materials once. You get this folder:

```
sql-data-warehouse-project/
├── 00_setup/            this guide, and the one-click Docker setup for Mac
├── datasets/
│   ├── source_crm/      cust_info.csv, prd_info.csv, sales_details.csv
│   └── source_erp/      CUST_AZ12.csv, LOC_A101.csv, PX_CAT_G1V2.csv
├── docs/                the architecture, the data flow, the data model, the catalog, the naming conventions
├── scripts/
│   ├── init_database.sql   creates the DataWarehouse database and its three schemas
│   ├── bronze/          ddl_bronze.sql, proc_load_bronze.sql
│   ├── silver/          ddl_silver.sql, proc_load_silver.sql
│   └── gold/            ddl_gold.sql
└── tests/               quality_checks_silver.sql, quality_checks_gold.sql
```

The project builds one database, `DataWarehouse`, with the schemas `bronze`, `silver` and `gold`. The scripts are the ones written in the videos, so you can type along or run them. The six CSV files are the two source systems, CRM and ERP, exactly as the videos use them.

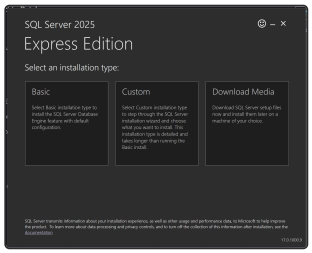
## 3 Windows: install SQL Server and SSMS

SQL Server Express is the free edition, and it is all you need. The videos show SQL Server 2022. The download page now offers 2025, and it works the same.

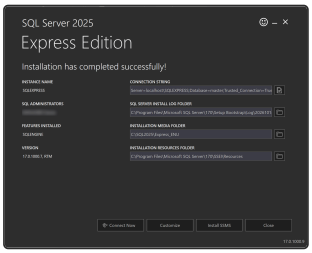
- Open the SQL Server downloads page, scroll to the **Express** card, and click **Download now**.
- Run the downloaded file and choose **Basic**. Accept the licence and click **Install**.
- The last screen shows your server name: `localhost\SQLEXPRESS`. Keep it, you need it in a minute. Your Windows account is now the administrator of this SQL Server.
- Download SQL Server Management Studio (SSMS) and install it with the default options. The installer may offer extra components. You need none of them.




Step 1: the Express card.




Step 2: choose Basic.

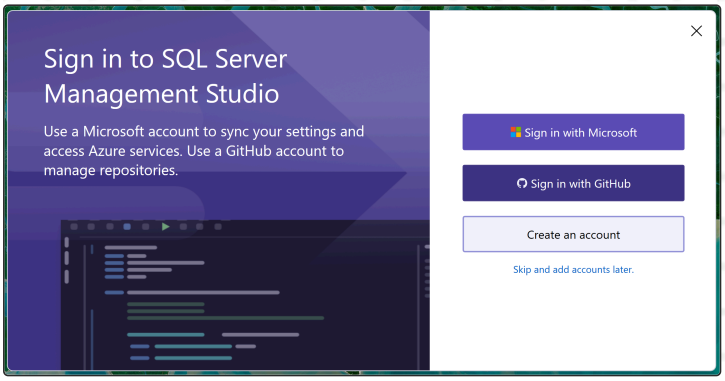



Step 3: the server name.

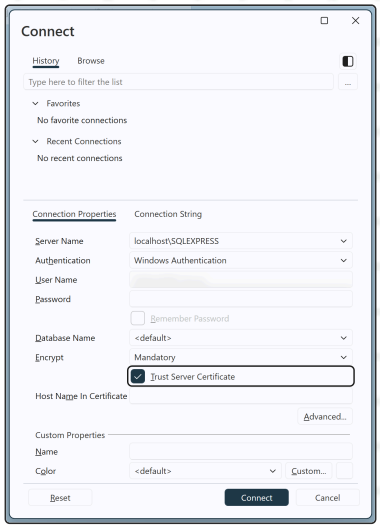
### Connect for the first time

SSMS 21 and newer look different from the videos.

- At the first start, SSMS asks you to sign in. You do not need an account: click **Skip and add accounts later** under the buttons.
- In the **Connect** window, type the server name `localhost\SQLEXPRESS`. If the field is empty, that is normal.
- Authentication: **Windows Authentication**.
- Tick **Trust Server Certificate**. Without it you get a certificate error.
- Click **Connect**. Your server appears on the left, in the Object Explorer.




Step 1: the skip link under the buttons.




Steps 2 to 4: the Connect window, filled in.

## 4 Windows: the files and the database

Two things before the first lesson of the project: the six files where the loader looks for them, and the empty database.

### Put the files where the scripts look

- Create the folder `C:\sql\dwh_project`.
- Copy the `datasets` folder into it. The files now sit at `C:\sql\dwh_project\datasets\source_crm\cust_info.csv` and so on: the paths written in `proc_load_bronze.sql`, the ones you see in the videos.
- A folder directly on `C:\` is readable by SQL Server. Downloads, Desktop and OneDrive often are not, and the load then fails with "Access is denied". If you keep the files somewhere else, change the six paths in `proc_load_bronze.sql` to your folder.

### Create the database

- In SSMS, click **File > Open > File**, open `scripts/init_database.sql`, and click **Execute** or press F5. It creates `DataWarehouse` with the three schemas. Refresh the Databases folder to see it.
- Every script after this one runs inside `DataWarehouse`. Before you execute, pick it in the database dropdown of the toolbar, where it says `master`. Otherwise the script stops with "schema name bronze does not exist".
- From here, follow the project: the bronze lesson creates the tables and the loader, and `EXEC bronze.load_bronze` fills them from the six files.

The script drops and recreates `DataWarehouse` every time it runs. Run it once, at the start. If you run it again later, your bronze, silver and gold work is gone.

## 5 Mac: Docker

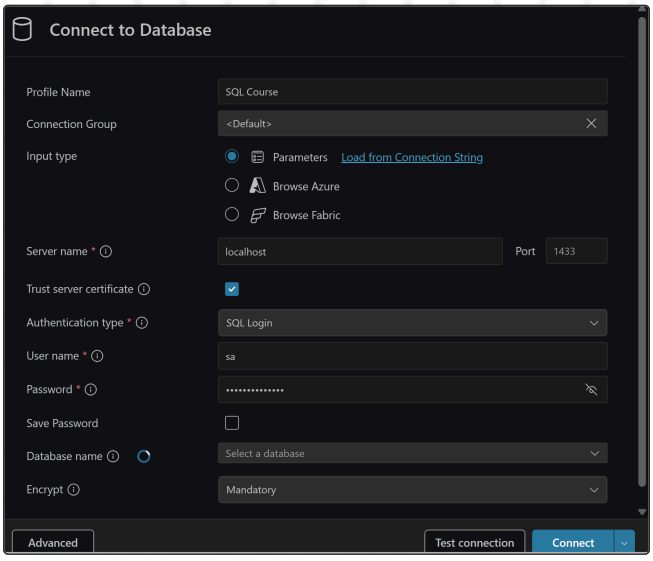
SQL Server does not install on macOS, and SSMS is Windows only. With Docker, SQL Server runs in a small container on your Mac, and you write your queries in Visual Studio Code. The SQL is the same, BULK INSERT and stored procedures included.

### Install, once

- Install Docker Desktop, the Apple chip version. Start it and leave it running.
- Docker Desktop settings, **General**. Set the Virtual Machine Manager to **Apple Virtualization framework** and turn on **Use Rosetta for x86_64/amd64 emulation on Apple Silicon**. SQL Server ships for Intel chips only, and this setting lets your Mac run it.
- Install Visual Studio Code and, inside it, the extension SQL Server (mssql) by Microsoft. It is Microsoft's tool for the Mac.

### Start SQL Server: one click

- In the project materials, open `00_setup/mac-docker` and double-click `start.command`.
- Wait a minute. The first start downloads SQL Server. Then the command puts the `datasets` folder inside SQL Server at `C:\sql\dwh_project\datasets\`, the path the loader uses. SQL Server on Linux reads `C:\` as its root folder, so the scripts run exactly as in the videos, nothing to edit.
- In Visual Studio Code, open the SQL Server extension on the left and add a connection. Server name `localhost`, port `1433`, tick **Trust server certificate**, authentication **SQL Login**, user `sa`, password `SqlServer!2026`.
- Open `scripts/init_database.sql` and run it. For every script after it, switch the database to `DataWarehouse`: click `master` in the bar at the bottom of the window and pick it. Then follow the project.




Step 3: the connection form, filled in.

### The same by hand

If you prefer your own commands: start SQL Server, then copy the datasets to the place the loader reads. Choose your own password: at least 8 characters with upper case, lower case, a number and a symbol.

```
docker run -e "ACCEPT_EULA=Y" -e "MSSQL_SA_PASSWORD=YourStrong!Passw0rd" \
  -p 1433:1433 --name sql --platform linux/amd64 \
  -d mcr.microsoft.com/mssql/server:2022-latest
docker exec -u root sql mkdir -p /sql/dwh_project
docker cp path/to/sql-data-warehouse-project/datasets sql:/sql/dwh_project/
docker exec -u root sql chown -R mssql /sql
```

Then connect as above. Microsoft's own guide for the first command: SQL Server in a Docker container.

> Did you run the SQL course package on this Mac? Its SQL Server is already there: the project's `start.command` finds the running container and uses it, with the same connection.

## 6 Windows inside your Mac: Parallels

This path gives you SSMS exactly as in the videos, because you run Windows on your Mac.

- Install Parallels Desktop. It downloads and installs Windows 11 for you in a few clicks. It costs about 100 dollars a year, with a 14-day free trial, and Windows needs its own licence.
- Give the virtual machine at least 6 GB of memory. A Mac with 16 GB is the comfortable minimum.
- Inside Windows, follow sections 3 and 4 like any Windows user.

SQL Server Express installs and runs on the ARM version of Windows 11, even though Microsoft does not list it as supported. Tested with SQL Server 2025 Express and SSMS 22 in Parallels on an Apple chip.

## 7 When something breaks

Find your message in the first column. The fix is in the last one.

### Install and connect

| What you see | Why | What to do |
|---|---|---|
| SSMS asks you to sign in at the first start | optional account | Click **Skip and add accounts later** |
| The server name field is empty | normal | Type `localhost\SQLEXPRESS` |
| Error 26, "Error Locating Server/Instance Specified" | SQL Server is not running, or the name is wrong | Open **Services**, start "SQL Server (SQLEXPRESS)". If you installed the Developer edition, the name is `localhost` without the instance |
| "The certificate chain was issued by an authority that is not trusted" | SSMS encrypts the connection | Tick **Trust Server Certificate** |
| Installer: "Wait on the Database Engine recovery handle failed" | SQL Server could not start during setup | Uninstall SQL Server, restart Windows, install again. If it fails again, the real reason is in the ERRORLOG file under `C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Log` |
| SQL Server 2025 setup stops at the rules check | installer problem on some machines | Install SQL Server 2022 Express instead. Everything here runs on it |
| Your laptop does not let you install anything | no administrator rights | SQL Server needs them. Ask IT, or use a personal computer |

### The database and the files

| What you see | Why | What to do |
|---|---|---|
| Msg 262, "CREATE DATABASE permission denied in database 'master'" | your Windows login is not an administrator of this SQL Server. Someone else installed it, often IT | Log in with the Windows account that installed SQL Server, or ask an administrator to add your login to the **sysadmin** role (Security > Logins > Properties > Server Roles) |
| Msg 2760, "The specified schema name 'bronze' either does not exist" | the query runs in `master`, not in `DataWarehouse` | Pick `DataWarehouse` in the database dropdown, then run the script again. If the database is missing, run `init_database.sql` first |
| Msg 4860, "Cannot bulk load. The file does not exist or you don't have file access rights" | the file is not at the path in the script | The loader reads `C:\sql\dwh_project\datasets\...`. Put the folder there, section 4, or change the six paths in `proc_load_bronze.sql` |
| Msg 4861, "Cannot bulk load because the file could not be opened. Operating system error code 5 (Access is denied.)" | SQL Server runs as a service and cannot read your folder | Move the folder to `C:\sql\dwh_project\`. Downloads, Desktop and OneDrive folders are the usual culprits |
| Msg 208, "Invalid object name 'bronze.crm_cust_info'" | the bronze tables are not created yet | Run `ddl_bronze.sql` in `DataWarehouse`, then the loader |
| Msg 4864 or 4832, a conversion error or an unexpected end of file during the bulk load | a CSV file was changed, usually by Excel saving it | Copy the original file from the package again. Open CSV files with a text editor, not with Excel |
| A .sql file opens in Notepad | the file is not linked to SSMS | Open SSMS first, then **File > Open > File** |

### Docker

| What you see | Why | What to do |
|---|---|---|
| "Cannot connect to the Docker daemon" | Docker Desktop is not running | Start Docker Desktop, wait for the whale icon, try again |
| "no matching manifest for linux/arm64" | the Intel image needs emulation | Turn on Rosetta in Docker Desktop, section 5. By hand, add `--platform linux/amd64` to the command |
| "port is already allocated" | another SQL Server listens on 1433, not one of these packages | Stop it, or change the first `1433` in `docker-compose.yml` to `1434` and connect to port `1434` |
| The container stops right after the start | the password breaks the rules | At least 8 characters with upper case, lower case, a number and a symbol |
| "SQL Server did not answer" from start.command | SQL Server could not start, most often Rosetta is off | Docker Desktop settings, **General**, turn on Rosetta, section 5. The container's log in Docker Desktop names the reason |

If you are still stuck, write in the Q&A with the exact error message and a screenshot of the window. Then I can see what happened.